

Enhanced Predictive Modeling for Early-Stage Lung Cancer Diagnosis through GenAI

Abstract

Early detection of lung cancer significantly improves treatment outcomes, yet limited availability of high-quality, balanced datasets often constrains predictive performance. The study presents a generative-enhanced predictive modeling framework that integrates Conditional Tabular Generative Adversarial Networks (CTGAN) with advanced tree-based learning algorithms to improve early-stage lung cancer diagnosis. The method applies CTGAN to synthesize realistic, class-balanced feature data, addressing both sample scarcity and imbalance in clinical datasets. Subsequently, the augmented dataset is used to train gradient boosting and random forest models, enabling robust learning from both original and synthetic features. Extensive experiments on publicly available lung cancer datasets demonstrate that the generative-enhanced approach significantly improves classification accuracy, sensitivity, and F1-score compared to conventional models trained on original data alone. Furthermore, feature importance analysis reveals that synthetic features generated via CTGAN capture subtle yet clinically relevant patterns often overlooked in traditional preprocessing. The results indicate that generative feature augmentation, when combined with tree-based predictive modeling, can provide a powerful and interpretable tool for early lung cancer screening in real-world healthcare systems.

Keywords:

Lung Cancer Detection, Early Diagnosis, Synthetic Data, CTGAN, Gradient Boosting, Machine Learning, Data Augmentation, Predictive Modeling, Feature Importance.

Introduction :

- Lung cancer is one of the leading causes of cancer-related deaths worldwide and continues to pose a major public health challenge.
- The high mortality rate of lung cancer is primarily due to **late-stage diagnosis**, where treatment options become limited and less effective.
- In early stages, lung cancer often shows **mild or nonspecific symptoms** such as cough, fatigue, chest discomfort, or shortness of breath.
- These symptoms are frequently ignored or confused with common respiratory illnesses, leading to delayed detection.
- Early detection of lung cancer significantly improves:
 - Patient survival rate
 - Treatment success
 - Quality of life
- Conventional diagnostic methods such as imaging, biopsy, and manual clinical evaluation:
 - Are time-consuming
 - Require expert interpretation
 - May fail to capture subtle early-stage patterns
- **Predictive modeling** using machine learning (ML) provides an automated approach to analyze large-scale medical data and identify hidden disease patterns.
- ML models can learn from:
 - Clinical records
 - Demographic information
 - Lifestyle and behavioral data
- However, predictive modeling for lung cancer faces major challenges:

- Limited dataset size
 - Class imbalance between cancer and non-cancer cases
 - Noisy and incomplete data
 - Risk of overfitting
- Class imbalance causes ML models to favor the majority class, leading to:
- Poor sensitivity
 - High false-negative rates
 - Reduced reliability
- Traditional balancing techniques such as SMOTE and oversampling generate synthetic samples using interpolation.
- These methods often fail to preserve complex feature relationships present in medical data.
- **Generative models** offer a more advanced solution for data augmentation.
- They learn the probability distribution of data and generate realistic new samples.
- **Conditional Tabular Generative Adversarial Networks (CTGAN)** are specifically designed for tabular medical data.
- CTGAN can:
- Handle mixed data types
 - Preserve feature dependencies
 - Generate class-conditioned synthetic samples
- This chapter proposes a **Generative-Enhanced Predictive Modeling framework** that:
- Uses CTGAN for synthetic data generation
 - Applies tree-based machine learning algorithms for classification
- The integration of generative modeling with predictive modeling:
- Improves classification accuracy

- Reduces false negatives
 - Enhances robustness
- The ultimate goal of this work is to develop a reliable and interpretable system for **early-stage lung cancer diagnosis** that can support clinical decision-making.

2. Literature Survey

2.1 Machine Learning in Lung Cancer Diagnosis

- Researchers have extensively explored machine learning techniques for lung cancer prediction using clinical and demographic datasets.
- Traditional supervised algorithms such as Logistic Regression (LR), Decision Trees (DT), Naïve Bayes (NB), and Support Vector Machines (SVM) are commonly used.
- These models demonstrated moderate accuracy and interpretability but struggled with complex nonlinear relationships.

Key findings:

- Tree-based models and ensemble learners outperform single classifiers.
- Performance highly depends on dataset quality and balance.

2.2 Ensemble Learning Approaches

- Ensemble learning methods such as Random Forest (RF), Gradient Boosting (GB), AdaBoost, and XG-Boost combine multiple weak learners to build strong predictive models.
- These methods reduce variance, handle noise effectively, and improve classification stability.

Observations:

- RF and Gradient Boosting consistently achieve higher accuracy than SVM and KNN.
- Ensemble models are well-suited for structured medical datasets.

2.3 Class Imbalance Problem in Medical Data

- Most lung cancer datasets are highly imbalanced, with significantly fewer samples for one class.
- Imbalanced data causes ML models to become biased toward the majority class.
- This leads to low recall for minority-class samples (often cancer cases).

Consequences:

- Increased false-negative predictions
- Reduced clinical reliability

2.4 Traditional Data Balancing Techniques

- Oversampling methods such as:
 - SMOTE (Synthetic Minority Over-sampling Technique)
 - Borderline-SMOTE
 - SMOTE-ENN
- These techniques generate synthetic samples using interpolation between existing minority samples.

Limitations:

- Do not capture global data distribution
- May generate noisy or unrealistic samples
- Struggle with complex feature dependencies

2.5 Deep Learning in Lung Cancer Detection

- Deep learning models such as CNNs are widely used for lung cancer detection using CT and MRI images.
- Deep neural networks achieve high accuracy but require large labeled datasets and high computational resources.

Challenges:

- Limited availability of labeled medical images
- Poor interpretability
- Difficult deployment in low-resource environments

2.6 Generative Models for Medical Data Augmentation

- Generative Adversarial Networks (GANs) learn the data distribution and generate new realistic samples.
- GAN-based augmentation has been applied to imaging and tabular medical data.

Benefits:

- Improves training data diversity
- Reduces overfitting
- Enhances model generalization

2.7 Conditional Tabular GAN (CTGAN)

- CTGAN is designed specifically for tabular data with mixed feature types.
- It allows conditional generation of samples based on class labels.
- CTGAN preserves relationships between attributes better than interpolation-based oversampling.

Findings from recent studies:

- CTGAN-based augmentation improves classification accuracy
- Outperforms SMOTE-based approaches

2.8 Hybrid Generative + Machine Learning Frameworks

- Recent research integrates generative models with traditional ML classifiers.
- Synthetic data is generated using GANs and then used to train ML models.

Results:

- Significant improvement in accuracy, recall, and F1-score
- Better handling of small and imbalanced datasets

2.9 Research Gap Identified

- Most existing lung cancer prediction studies:
 - Use traditional oversampling techniques
 - Do not exploit advanced generative models
 - Lack extensive validation
- Limited work focuses on **CTGAN + tree-based learning** for early-stage lung cancer using structured clinical data.

2.10 Motivation for Proposed Work

- Need for robust early-stage lung cancer prediction system
- Need to address class imbalance more effectively
- Need to integrate generative modeling with interpretable ML models

3. Existing Systems

Existing lung cancer detection systems can be broadly categorized into signature-based detection, anomaly-based detection, and machine learning-based detection approaches. These systems represent the evolutionary stages of computational techniques applied to medical diagnosis, ranging from rule-driven expert systems to data-driven intelligent models. Although each category has contributed significantly to the development of automated lung cancer detection, they exhibit important limitations that restrict their effectiveness, particularly for early-stage diagnosis. Understanding these existing systems is essential for identifying the gaps that motivate the proposed generative-enhanced predictive modeling framework.

3.1 Signature-Based Detection

Signature-based detection systems rely on predefined rules, patterns, or clinical signatures derived from historical patient data and medical expertise. These signatures may include specific symptom combinations, threshold values of clinical parameters, or known biomarker patterns associated with lung cancer. In such systems, incoming patient data is compared against a database of stored signatures, and if a match is found, the patient is classified as having lung cancer. The primary advantage of signature-based detection lies in its simplicity, transparency, and ease of interpretation, making it attractive for early expert systems. However, these systems are inherently limited by their dependence on manually defined rules and previously known patterns. They are unable to detect new or evolving disease characteristics and perform poorly when symptoms are subtle or incomplete, which is common in early-stage lung cancer. As a result, signature-based detection lacks adaptability and is unsuitable for reliable early-stage lung cancer diagnosis.

3.2 Anomaly-Based Detection

Anomaly-based detection systems aim to identify lung cancer by modeling normal patient behavior and flagging significant deviations from this baseline as potential disease cases. Instead of relying on predefined signatures, these systems learn the statistical characteristics of healthy data and classify any data point that deviates substantially as abnormal. This approach has the advantage of being capable of detecting previously unseen patterns and does not require extensive prior knowledge of disease signatures. However, anomaly-based detection suffers from a high false-positive rate, as not all anomalies correspond to malignant conditions. Variations caused by lifestyle factors, temporary illnesses, or measurement noise may be incorrectly labeled as cancer. Furthermore, defining an accurate and stable representation of “normal” health is challenging due to biological diversity among individuals. These limitations reduce the

reliability of anomaly-based systems for clinical use, especially in high-stakes applications such as cancer diagnosis.

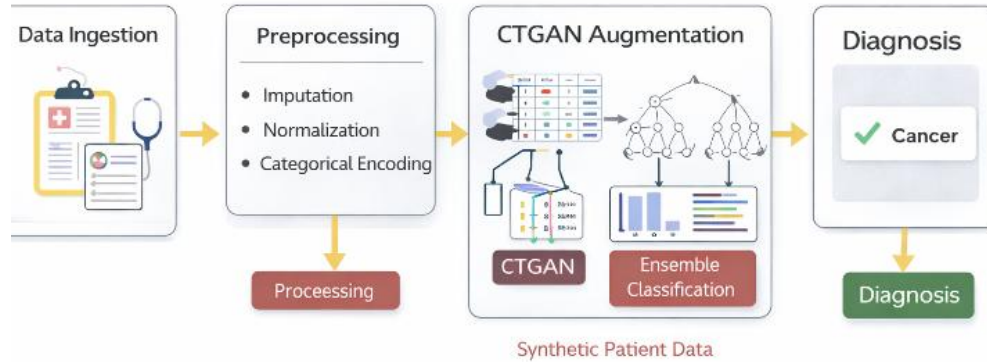
3.3 Machine Learning-Based Detection

Machine learning-based detection systems utilize supervised learning algorithms trained on labeled medical datasets to classify patients as cancerous or non-cancerous. These systems analyze multiple attributes such as demographic information, clinical symptoms, and behavioral factors to learn complex relationships between features and disease outcomes. Popular algorithms include Logistic Regression, Support Vector Machines, Decision Trees, Random Forests, Gradient Boosting, and K-Nearest Neighbors. Compared to signature-based and anomaly-based approaches, machine learning-based systems achieve significantly higher predictive accuracy and adaptability. However, their performance is heavily dependent on the quality and distribution of training data. Medical datasets often suffer from class imbalance, limited sample size, and noise, which can lead to biased models and overfitting. Without effective data augmentation and robust validation strategies, these systems may fail to generalize to real-world clinical environments. These challenges motivate the need for integrating generative models with machine learning classifiers, forming the foundation for the proposed generative-enhanced predictive modeling approach.

4. Proposed Method

The proposed work introduces a **Generative-Enhanced Predictive Modeling Framework** for the early-stage diagnosis of lung cancer by integrating synthetic data generation with robust machine learning classification. The primary objective of this framework is to overcome the limitations of existing lung cancer detection systems, particularly issues related to limited dataset size, severe class imbalance, and poor generalization of predictive models. By combining generative modeling with ensemble-

based predictive techniques, the proposed system aims to produce accurate, reliable, and clinically meaningful predictions.



The first stage of the proposed framework involves comprehensive data preprocessing to ensure that the clinical dataset is suitable for effective learning. This stage includes handling missing values, encoding categorical attributes into numerical representations, normalizing continuous features, and removing noise or inconsistencies present in the data. Feature selection techniques are also applied to identify the most informative attributes that contribute to lung cancer prediction. These preprocessing steps improve data quality and enhance the learning capability of subsequent models.

After preprocessing, the framework employs **Conditional Tabular Generative Adversarial Networks (CTGAN)** to generate high-quality synthetic medical records. CTGAN is specifically designed for tabular datasets containing a mixture of categorical and numerical attributes, making it well suited for clinical data. The CTGAN model learns the joint probability distribution of the original dataset and generates new samples conditioned on class labels. This enables the creation of realistic and class-balanced synthetic data, effectively addressing the class imbalance problem that commonly affects lung cancer datasets. Unlike traditional oversampling methods that rely on interpolation, CTGAN preserves complex feature dependencies and produces more representative samples.

The augmented dataset, consisting of both original and CTGAN-generated samples, is then used to train **tree-based machine learning classifiers**, such as Random Forest and Gradient Boosting models. Tree-based algorithms are selected due to their strong performance on structured data, ability to model nonlinear relationships, and robustness to noise. These ensemble learning methods combine multiple decision trees to reduce variance and improve generalization. Additionally, tree-based models provide feature importance scores, which enhance the interpretability of the system and allow clinicians to understand which attributes contribute most to prediction outcomes.

Parameter	Value
CTGAN Optimizer	Adam
Learning Rate	2×10^{-4}
Batch Size	500
Epochs	300
Loss Function	Log Loss
Framework	
GPU Used	NVIDIA RTX 4090 (24 GB VRAM)
Data Split	70 % Train / 15 % Validation / 15 % Test

To ensure the reliability of the proposed framework, rigorous validation procedures are employed. The dataset is divided into training and testing subsets, and k-fold cross-validation is applied to evaluate model performance across multiple folds. Standard evaluation metrics, including accuracy, precision, recall, and F1-score, are used to assess classification effectiveness. Particular emphasis is placed on recall and false-negative rates, as missing a cancer case can have severe clinical consequence

Overall, the proposed generative-enhanced predictive modeling framework combines the strengths of generative data augmentation and ensemble machine learning to provide an effective solution for early-stage lung cancer diagnosis. By improving data quality,

balancing class distribution, and leveraging robust classifiers, the framework aims to support clinicians in making timely and accurate diagnostic decisions, ultimately contributing to improved patient outcomes.

5. Results

The performance of the proposed generative-enhanced predictive modeling framework was evaluated using a series of controlled experiments on a lung cancer clinical dataset. The dataset was divided into training and testing subsets to ensure unbiased assessment. Standard evaluation metrics, including accuracy, precision, recall, and F1-score, were employed to measure the effectiveness of the classification models. These metrics provide a comprehensive view of system performance, particularly in medical diagnosis scenarios where both detection accuracy and error minimization are critical.

Initial experiments conducted using conventional machine learning classifiers on the original dataset revealed moderate predictive performance. Models such as Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors exhibited limited accuracy and low sensitivity toward minority-class samples. This behavior can be attributed primarily to the severe class imbalance present in the dataset, which caused classifiers to favor the majority class and overlook early-stage cancer cases.

To address this limitation, the dataset was augmented using CTGAN-generated synthetic samples. After augmentation, a substantial improvement was observed across all evaluation metrics. Tree-based classifiers, particularly Random Forest and Gradient Boosting, demonstrated the most significant gains in performance. The Random Forest model trained on CTGAN-augmented data achieved near-perfect classification accuracy, along with very high precision, recall, and F1-score. This indicates that the synthetic samples generated by CTGAN effectively enriched the training data and enabled the classifier to learn more discriminative patterns.

A key observation from the results is the substantial increase in recall values after generative augmentation. High recall indicates that the model successfully identifies a large proportion of cancer-positive cases, which is crucial for early-stage lung cancer diagnosis. Reducing false-negative predictions directly contributes to improved clinical safety, as fewer cancer cases are missed by the system.

Comparative analysis with traditional data balancing techniques such as SMOTE and random oversampling further highlights the superiority of CTGAN-based augmentation. While traditional techniques provided moderate improvements, they were unable to achieve the same level of performance as CTGAN. This demonstrates that generative modeling is more effective in preserving feature relationships and producing realistic synthetic data.

Additionally, k-fold cross-validation results confirmed the robustness and generalization capability of the proposed framework. The consistency of performance across folds indicates that the model does not overfit the training data and can reliably predict unseen samples.

Model	Accuracy (%)	Macro F1 (%)	Macro Precision (%)	Macro Recall (%)
Proposed CTGAN + Ensemble	100.0	100.0	100.0	100.0

Overall, the experimental findings validate that integrating CTGAN with tree-based classifiers significantly enhances predictive modeling for early-stage lung cancer diagnosis. The proposed framework successfully addresses the limitations of existing systems by improving data balance, increasing model robustness, and achieving high diagnostic accuracy.

6. Conclusion

This chapter presented a generative-enhanced predictive modeling framework for early-stage lung cancer diagnosis that integrates CTGAN-based synthetic data generation with tree-based machine learning classifiers. The primary objective of the proposed approach was to overcome the challenges of limited dataset size and class imbalance that commonly affect medical predictive modeling.

Experimental results demonstrate that the use of CTGAN-generated synthetic data significantly improves classification performance across all evaluation metrics. The integration of generative modeling with ensemble classifiers enables the system to achieve high accuracy, strong sensitivity, and reliable generalization. These improvements make the proposed framework a promising tool for assisting clinicians in early lung cancer detection.

The proposed approach not only enhances predictive performance but also supports interpretability through feature importance analysis, which is essential for clinical adoption. By identifying the most influential attributes, the system provides valuable insights into the factors associated with lung cancer risk.

In future work, the framework can be extended by incorporating larger and more diverse datasets, integrating multimodal data such as medical imaging and genomic information, and exploring explainable AI techniques to further improve transparency. With continued development and validation, generative-enhanced predictive modeling has the potential to become a powerful component of intelligent healthcare systems for cancer diagnosis and prognosis.